FUTURESTACK HACKATHON · ROUND 2

Rakshak 2.0

A working brand-governance review desk — paste a draft, get rule-cited findings with fixes, export clean text + audit report. 100% local-first.

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Live prototype: hariharan-007design.github.io/Rakshak2.0
Repository: github.com/hariharan-007design/Rakshak2.0

Results & Proof WHAT WORKS NOW

A judge can open the live link and complete the full workflow without help: paste a draft → run a brand check → see findings with rule citations → accept/ignore each → download a report. Everything runs in the browser.

3	40+	<1s	100%
Rule packs (Mozilla, Mailchimp, Acme)	Deterministic rule checks	Scan time on a draft	Local-first (no cloud required)

Seeded test — measurable protocol

Metric	Target	Result	How measured
Violations detected (recall)	≥ 11 / 12	**12 / 12**	Seeded set: 4 violations per pack, each mapped to one rule ID
False discovery proportion	< 10%	**0 / 20**	20 clean control passages, zero spurious findings
Deterministic scan speed	< 2 s / asset	**< 1 s**	Local timer on a representative draft
Offline operation	0 content network calls	**Verified**	Deterministic scan runs with no network; only optional AI polish calls an external API

> **Every finding is grounded.** Each one cites a specific rule ID and the guideline source URL — no "AI thinks". Human decisions (accept / ignore / edit) are recorded in the exported audit log.

What We Built — The Review Loop

Paste draft / upload .txt .md .pdf → Select company rule pack → Scan (deterministic + optional AI) → Findings with fixes → Accept / ignore each → Revised draft + report

Three demo workspaces

Workspace	Rule pack source	Purpose
Mozilla Demo	mozilla.org/brand + Trademark Policy	Tone, terminology, trademark-attribution, implied-endorsement
Mailchimp Demo	styleguide.mailchimp.com	Voice, accessibility, CAN-SPAM footer
Acme SaaS (synthetic)	Neutral placeholder, no affiliation	Seeded violations + clean control passages

Logo review (Round 2): upload a logo file and get file-quality checks — dimensions, transparency sampling, clear-space heuristic — labelled as *preliminary checks, not brand approval*.

Optional AI polish: a BYO-key assistant (Groq/OpenAI-compatible) rewrites a flagged sentence. The key stays in the browser; the deterministic rules remain the source of truth. Output is clearly marked *AI-generated · review before use* and never silently replaces the reviewed draft.

Agent Architecture (MVP Scope)

What "agent" means here: the review loop — it selects relevant rules, runs deterministic and contextual checks, produces structured findings, proposes fixes, records human decisions, and (optionally) uses accepted terminology preferences in later scans. Autonomy is intentionally limited to preserve human control.

Check types — reliability is surfaced, not hidden

Type	Examples	Reliability	UI badge
Deterministic	Banned phrases, required disclaimers, exact terminology, color hex, date/cert expiry	High — exact match against a configured rule	**Rule match**
Advisory (LLM+RAG)	Tone alignment, clarity, inclusivity, fair-language heuristics	Context-dependent; can miss / invent	**AI assessment — verify**

> **Honest boundary:** the tool never says "compliant" or "approved". It only reports "no deterministic violations found; X advisory items flagged for review". It is a pre-screening aid, not a compliance guarantee.

Technology

Layer	Choice	Rationale
Prototype	Static web app (HTML / JS / CSS)	Runs on any judge's machine through a link — zero install, zero backend
Document handling	PDF.js (bundled)	Extract selectable text from .pdf in-browser; explicit size/page limits surfaced
Deterministic checks	Regex + JS	High-confidence for explicit rules; fast and predictable
Findings / fixes	Structured objects with rule ID + source URL	Traceable, human-controlled corrections
Diff / export	diff-match-patch, PDF	Side-by-side revised draft + downloadable report
Optional AI	OpenAI-compatible API (Groq), BYO key	Reliable rewrite assistant; deterministic layer stays local
Storage	localStorage (session only)	No account, no server; data not saved after tab closes

Original Round 1 target stack (Tauri v2 + Ollama + sqlite-vec) remains the post-hackathon desktop plan. For Round 2 the browser prototype was chosen so the prototype is directly usable by judges on any device.

Existing Solution Comparison

Solution	Checks at creation	User-owned rules	Local / private	Cited findings	Human-controlled fixes
Brand portals (Bynder, Frontify)	No	Partial	No	No	No
AI writing (Writer, Jasper)	No	Partial	Limited	Often weak	Partial
Custom GPT	Manual	Manual	No	Unreliable	No
MLR workflow (Veeva)	No — post-creation	Yes	Enterprise-dependent	Partial	Yes
Rakshak 2.0 (MVP)	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

Known Limitations (Honest Boundaries)

- **Language:** English only in this build.
- **Rule quality:** demo packs are hand-curated; production PDF→JSON extraction is post-hackathon.
- **Scanned PDFs:** not supported in MVP (no OCR yet); selectable-text PDFs work.
- **Logo checks:** file-quality heuristics only — not approval of an approved brand asset.
- **AI polish:** optional and BYO-key; a fluent rewrite can still violate a rule, so output is re-checkable and human-reviewed.
- **Scope:** pre-screening aid only — does not replace MLR, legal, regulatory, or accessibility approval.

Future Scope (Post-Hackathon)

Phase	Scope	Trigger
v0.2	VS Code / Figma / Chrome plugins (same agent core)	"Can it work in my tool?"
v0.3	Encrypted team vault (Syncthing / iCloud / Drive, no server)	Small teams want shared memory
v1.0	Optional SaaS dashboard for admins (opt-in)	Enterprise asks for fleet visibility
Community packs	Shareable rule packs: "Pharma MLR", "Chem GHS/REACH", "Fintech FINRA"	Local-first, privacy-by-default; users contribute back

Three meaningful Round 3 improvements

1. **Multimodal document & asset review** — OCR, page-level evidence, approved-logo matching, accessibility checks. *Proof:* a scanned document produces a finding linked to the exact page/region.
2. **Enterprise governance & audit** — real auth, role-based permissions, isolated workspaces, versioned rules, auditable approval history. *Proof:* an unauthorized user cannot access another company's workspace.
3. **Controlled publishing & quality measurement** — approval-before-publishing, audit trails, false-positive / reviewer-override metrics. *Proof:* unapproved content cannot publish.

Team & Participant Details

Field	Value
Team Name	ssshariharan82
Project Name	Rakshak 2.0
Team Leader	Hariharan S
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Team Members	Hariharan S
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Why us: Chemical-engineering background (GHS/SDS/regulatory rigor) + hackathon project experience (dashboard / workflow / system design) = domain credibility for future regulated packs and execution capability for this build.